CAMBRIDGE
International Education

Cambridge International AS & A Level

MATHEMATICS **9709/51**

Paper 5 Probability & Statistics 1 **October/November 2024**

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:
Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:
Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:
Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:
Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:
Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Mathematics-Specific Marking Principles

1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.

2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.

3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.

4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).

5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.

6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

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Mark Scheme Notes

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

DM or **DB** When a part of a question has two or more 'method' steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

FT Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.

- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
- For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
- The total number of marks available for each question is shown at the bottom of the Marks column.
- Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
- Square brackets [ ] around text or numbers show extra information not needed for the mark to be awarded.

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Abbreviations

AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no 'follow through' from a previous error is allowed)
CWO	Correct Working Only
ISW	Ignore Subsequent Working
SOI	Seen Or Implied
SC	Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)
WWW	Without Wrong Working
AWRT	Answer Which Rounds To

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Question	Answer	Marks	Guidance
1(a)	**Method 1**		
	[P($X < 8$) =] $1-\left(\frac{5}{6}\right)^7$	**M1**	$1 - b^d$, $b = \frac{5}{6}, \frac{1}{6}$ $d = 7, 8$. $1 - c^e - (1 - c) \times c^{e-1}$, $c = \frac{5}{6}, \frac{1}{6}$ $e = 8,9$.
	= 0.721	**A1**	0.720918… $\frac{201811}{279936}$. If M0 scored, **SC B1** for 0.7209… or $\frac{201811}{279936}$ only.
	Method 2		
	[P($X < 8$) =] $\frac{1}{6}+\left(\frac{5}{6}\right)\left(\frac{1}{6}\right)+\left(\frac{5}{6}\right)^2\left(\frac{1}{6}\right)+\left(\frac{5}{6}\right)^3\left(\frac{1}{6}\right)+\left(\frac{5}{6}\right)^4\left(\frac{1}{6}\right)+\left(\frac{5}{6}\right)^5\left(\frac{1}{6}\right)+\left(\frac{5}{6}\right)^6\left(\frac{1}{6}\right)$	**M1**	$a+ba+b^2a+b^3a+b^4a+b^5a+b^6a\left[+b^7a\right]$. $a=\frac{1}{6}, \frac{5}{6}a+b=1$.
	= 0.721	**A1**	0.720918… $\frac{201811}{279936}$. If M0 scored, **SC B1** for 0.7209… or $\frac{201811}{279936}$ only.
		2	

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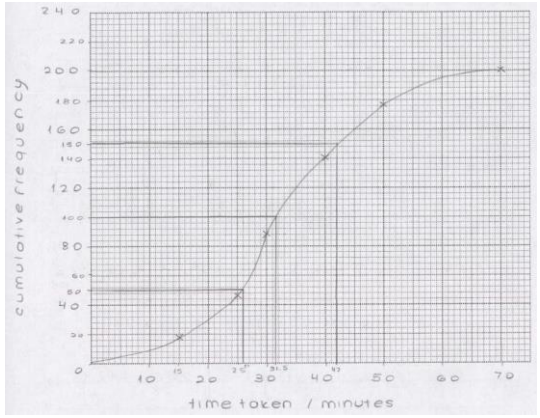
Question	Answer	Marks	Guidance
1(b)	$\left(\frac{5}{6}\right)^6 \times \left(\frac{1}{6}\right)^2 \times 7$	**M1**	$\left(\frac{5}{6}\right)^6 \times \left(\frac{1}{6}\right)^2 \times d$ d an integer ≥ 1, no inappropriate addition.
	0.0651	**A1**	$0.0651 \leqslant p < 0.06512$.
		2	

Question	Answer	Marks	Guidance
2(a)	[Probs $6k, 3k, 2k, 6k, 11k$ so $28k = 1$,] $k = \frac{1}{28}$	**B1**	k must be identified
	x: −2, −1, 0, 2, 3 P($X = x$): $\frac{6}{28}$ 0.2143; $\frac{3}{28}$ 0.1071; $\frac{2}{28}$ 0.07143; $\frac{6}{28}$ 0.2143; $\frac{11}{28}$ 0.3929	**M1**	Table with correct outcomes and 2 correct probabilities. FT substituting *their k* correctly into formula, with $0 < p < 1$. No additional x values unless probability 0. Condone in terms of k of the form $\frac{6k}{28}$ or $6k$.
		A1	Fully correct. Decimal answers to at least 3 sig figures, condone not summing exactly to 1.
		3	

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Question	Answer	Marks	Guidance
2(b)	$\mathrm{E}(X) = \left[-2\times\frac{6}{28}+-1\times\frac{3}{28}+\left[0\times\frac{2}{28}\right]+2\times\frac{6}{28}+3\times\frac{11}{28}\right]$ $\frac{1}{28}(-12-3+12+33)\left[=\frac{15}{14}\right]$	**M1**	Accept unsimplified expression. May be calculated in the variance. FT *their* table with 5 probabilities $0 < p < 1$ that sum to 1.
	$\mathrm{Var(X)} = \frac{6\times(-2)^2+3\times(-1)^2+6\times2^2+11\times3^2}{28}$ $-their\left(\frac{15}{14}\right)^2$	**M1**	Appropriate variance formula using *their* $(\mathrm{E}(X))^2$ value. FT *their* table with at least 4 probabilities $0 < p < 1$, that may not sum to 1.
	$= 4.21,\ 4\frac{41}{196}$	**A1**	Condone $\frac{825}{196}$. If one or both M marks not awarded, **SC B1** for correct answer WWW.
		3	

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Question	Answer	Marks	Guidance
3(a)		**M1**	At least 4 points plotted within tolerance at upper bounds. Linear cf scale $0 \leqslant cf \leqslant 200$ and linear time scale $0 \leqslant t \leqslant 70$, with at least 3 values identified on each. Minimum scale uses at least ½ the grid.
		A1	All points plotted correctly. Curve drawn and joined to (0, 0). Axes labelled cumulative frequency (cf), time (t) and minutes (min) – or a suitable title.
		2	
3(b)	Median = 33	**B1 FT**	Must be identified. Evidence of use of graph must be seen. Strict FT ± ½ square on time axis.
	[IQR =] 42 – 26	**M1**	$41 \leqslant UQ \leqslant 43 - 25 < LQ \leqslant 27$. If outside of range FT ± ½ square on time axis.
	16	**A1 FT**	
		3	

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Question	Answer	Marks	Guidance
3(c)	<table><tr><td>Midpoint</td><td>7.5</td><td>20</td><td>27.5</td><td>35</td><td>45</td><td>60</td></tr><tr><td>Frequency</td><td>18</td><td>28</td><td>42</td><td>52</td><td>36</td><td>24</td></tr></table>	**B1**	At least 5 correct midpoints or 5 correct frequencies seen.
	Mean = $\dfrac{18\times 7.5+28\times 20+42\times 27.5+52\times 35+36\times 45+24\times 60}{200}$	**M1**	Correct mean formula using *their* 6 midpoints (must be within class, not upper bound, not lower bound) condone 1 error and *their* 6 frequencies (not cumulative frequencies).
	$= 33.65,\ 33\dfrac{13}{20}$	**A1**	Accept 33.7, not $\dfrac{673}{20}$.
		3	

Question	Answer	Marks	Guidance
4(a)	P(HRR) = $\dfrac{1}{4}\times\dfrac{4}{6}\times\dfrac{4}{6}=\dfrac{16}{144},\dfrac{4}{36}$ P(TRR) = $\dfrac{3}{4}\times\dfrac{3}{7}\times\dfrac{2}{6}=\dfrac{18}{168},\dfrac{3}{28}$ P(HBB) = $\dfrac{1}{4}\times\dfrac{2}{6}\times\dfrac{2}{6}=\dfrac{4}{144},\dfrac{1}{36}$ P(TBB) = $\dfrac{3}{4}\times\dfrac{4}{7}\times\dfrac{3}{6}=\dfrac{36}{168},\dfrac{6}{28},\dfrac{3}{14}$	**B1**	2 clearly identified unsimplified probabilities from P(HRR), P(TRR), P(HBB) and P(TBB) correct.
	$\dfrac{4}{36}+\dfrac{3}{28}+\dfrac{1}{36}+\dfrac{6}{28}$	**M1**	Sum of 4 correct scenarios, may be identified by the unsimplified probability calculations.
	$=\dfrac{29}{63}$ or 0.460	**A1**	(0.460317… to at least 3SF).
		3	

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Question	Answer	Marks	Guidance
4(b)	$\left[\mathrm{P(T\|BB)} = \dfrac{\mathrm{P(T} \cap \mathrm{BB)}}{\mathrm{P(BB)}}\right] = \left[\dfrac{\frac{3}{4}\times\frac{4}{7}\times\frac{3}{6}}{\frac{1}{36}+\frac{6}{28}}\right]$ $\frac{3}{14} \div \frac{61}{252}$ or $\dfrac{\frac{3}{14}}{\frac{61}{252}}$	**M1**	$\frac{3}{4}\times\frac{4}{7}\times\frac{3}{6}, \frac{36}{168}$ oe, $\frac{3}{14}, 0.2142857\ldots$ seen as numerator of a fraction, accept unsimplified, FT *their* P(TBB) from **4(a)**
		M1	*their* $\frac{1}{36} + $ *their* $\frac{6}{28}$ FT from **4(a)** or correct, 0.24206…, seen as denominator of a fraction, accept unsimplified.
	$\frac{54}{61}$ or 0.885	**A1**	Accept 0.8852589… rounded to at least 3SF. If one or both Ms not awarded, **SC B1** for correct final answer WWW.
		3	

Question	Answer	Marks	Guidance
5(a)	$\mathrm{P}(83 < X < 95) = \mathrm{P}(\frac{83-90}{8} < Z < \frac{95-90}{8})$	**M1**	Using ± standardisation formula with 90, 8 and *either* 83 *or* 95. Not σ^2 , not σ, no continuity correction.
	$= \mathrm{P}(-0.875 < Z < 0.625)$	**A1**	Both ±0.875 OE and ±0.625 OE seen. If M0 scored, **SC B1** for both ±0.875 and ±0.625 seen
	$[\Phi(0.625) + \Phi(0.875) - 1]$ $= 0.7340 + 0.8092 - 1$	**M1**	Calculating the appropriate probability area, leading to their final probability. Expect final answer > 0.5.
	$= 0.543$	**A1**	0.5432, $0.543 \leqslant p < 0.5435$. Only dependent on the 2nd M mark.
		4	

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Question	Answer	Marks	Guidance
5(b)	[Mean $=160\times0.6=]96$ [Var $=160\times0.6\times0.4=]38.4$	**B1**	96 and 38.4 seen, allow unsimplified. May be seen in the standardisation formula. $\left(\begin{array}{l}\frac{8\sqrt{15}}{5}, 6.19677\ldots \text{to at least 4 SF}\\ \text{implies correct variance}\end{array}\right)$ Withold mark if variance clearly identified as standard deviation, condone N(96, $\sqrt{38.4}$) if standardisation formula correct or variance/standard deviation correctly stated as well.
	P(X < 105) = P($Z < \frac{104.5-96}{\sqrt{38.4}}$) [P($Z < 1.372$) = $\Phi(1.372)$]	**M1**	Substituting *their* 96 and *their* 38.4 into the ±standardising formula (any number for 104.5), condone σ^2 or $\sqrt{\sigma}$.
		M1	Use continuity correction 104.5 or 105.5 in *their* standardisation formula. Note: $\frac{\pm8.5}{\sqrt{38.4}}$ or $\frac{\pm8.5}{6.197}$ seen gains M2 BOD.
		M1	Appropriate area Φ, from final process, must be a probability. Expect final answer > 0.5.
	= 0.915[0]	**A1**	$0.9149 \leqslant p \leqslant 0.915$. If one or more M marks not scored, **SC B1** for $0.9149 \leqslant p \leqslant 0.915$.

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Question	Answer	Marks	Guidance
6(a)	$\frac{182.7-\mu}{\sigma}=1.282$ $\frac{162.5-\mu}{\sigma}=-0.253$	**B1**	1.282 or – 1.282 seen, CAO (critical value).
		B1	$-0.2535 < z \leqslant -0.253$ or $0.253 \leqslant z < 0.2535$ seen.
		M1	One standardisation formula, not σ^2, or $\sqrt{\sigma}$, with 182.7 or 162.5 substituted correctly equated to a z value (not 0.9, 0.1, 0.8159, 0.5398, 0.4, 0.6, 0.6554, 0.7257, …).
	Solve, obtaining values for μ and σ	**M1**	Either a single expression with one variable eliminated formed or two expressions with both variables on the same side seen with at least one variable value stated.
	$\mu=165.8,\ \ \sigma=13.2$	**A1**	Answers must be to at least 1 DP (context).
		5	

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Question	Answer	Marks	Guidance
6(b)	**Method 1**		
	[P(X < 8) = 1 – P(8, 9, 10) =] 1 – ($^{10}C_8 (0.6)^8 (0.4)^2 + {}^{10}C_9 (0.6)^9 (0.4)^1 + (0.6)^{10}$) [= 1 – (0.12093 + 0.040311 + 0.0060466)]	**M1**	One term $^{10}C_x (p)^x (1-p)^{10-x}$. With $0 < p < 1, x \neq 0$ or 10.
		A1	Correct unsimplified expression. Allow 10 for $^{10}C_9$. Condone omission of last bracket only. If both brackets omitted in unsimplified expression allow recovery for final stated calculation of 1 – 0.1673 or final answer WRT 0.8327.
	= 0.833	**B1**	0.8327 < $p \leqslant$ 0.833.
	Method 2		
	$(0.4)^{10} + {}^{10}C_1 (0.6)^1 (0.4)^9 + {}^{10}C_2 (0.6)^2 (0.4)^8 + {}^{10}C_3 (0.6)^3 (0.4)^7 + {}^{10}C_4 (0.6)^4 (0.4)^6 + {}^{10}C_5 (0.6)^5 (0.4)^5 + {}^{10}C_6 (0.6)^6 (0.4)^4 + {}^{10}C_7 (0.6)^7 (0.4)^3$ $\left[1.0486\times10^{-4} + 1.5729\times10^{-3} + \ldots + 0.21499\right]$	**M1**	One term $^{10}C_x (p)^x (1-p)^{10-x}$. With $0 < p < 1, x \neq 0$ or 10.
		A1	Correct unsimplified expression.
	= 0.833	**B1**	0.8327 < $p \leqslant$ 0.833.
		3	

Question	Answer	Marks	Guidance
7(a)	$\frac{8!}{2!2!}$	**M1**	$\frac{k!}{2!m!}$ $k = 7$ or 8, $m = 1, 2$.
	= 10080	**A1**	
7(b)	**Method 1** Number of ways with no restriction on Es – ways with Es together		
	$\frac{7!}{2!2!} - \frac{6!}{2!}$	**M1**	$\frac{7!}{2!2!}$ $- r$, r integer > 1.
	[= 1260 – 360]	**M1**	$s - \frac{6!}{2!}$, s integer > 360.
	= 900	**A1**	
	Method 2 T ^ ^ ^ ^ ^ T with Es inserted in gaps		
	$\frac{5!}{2!} \times \frac{6 \times 5}{2}$ or $\frac{5!}{2!} \times {}^6C_2$	**M1**	$t \times \frac{6 \times 5}{2}$ or $t \times {}^6C_2$, t an integer > 1.
	[= 60 × 15]	**M1**	$\frac{5!}{2!} \times u$, u an integer > 1.
	=900	**A1**	
		3	

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Question	Answer	Marks	Guidance
7(c)	**Method 1 – addition**		
	T E _ _ = ${}^2C_1 \times {}^2C_1 \times {}^5C_2 = 40$	**B1**	Either identified or correct unsimplified expression, either alone or in an addition.
	T E E _ = ${}^2C_1 \times {}^2C_2 \times {}^5C_1 = 10$	**B1**	Either identified or correct unsimplified expression, either alone or in an addition.
	Probability $\frac{(40+10)}{{}^9C_4}$	**M1**	$\frac{a}{{}^9C_4}$, a an integer < 126. Denominator value must be seen as ${}^9C_4$ somewhere.
	$\left[\text{Percentage} = \frac{50}{126} \times 100\right] = 39.7\%$	**A1**	39.68 ⩽ percentage ⩽ 39.7.
	Method 2 – subtraction (total arrangements with 1 T – number of arrangements with 1T 0 E)		
	T ^ ^ ^ = ${}^2C_1 \times {}^7C_3 = 70$	**B1**	Either identified or correct unsimplified expression, either alone or in a subtraction.
	T * * * = ${}^2C_1 \times {}^5C_3 = 20$	**B1**	Either identified or correct unsimplified expression, either alone or in a subtraction.
	Probability $\frac{(70-20)}{{}^9C_4}$	**M1**	$\frac{a}{{}^9C_4}$, a an integer < 126. Denominator value must be seen as ${}^9C_4$ somewhere.
	$\left[\text{Percentage} = \frac{50}{126} \times 100\right] = 39.7\%$	**A1**	39.68 ⩽ percentage ⩽ 39.7.
		4	

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